

Linear Discriminant Analysis

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Recap

- Last time we covered ***k*-Means Clustering**:
 - Algorithm.
 - Selecting the number of groups.
 - Fuzzy clustering.

Introduction

- Next, we are moving into classification.
- How are these approaching similar and different to clustering?
- We can classify objects according to multiple objectives:
 - Descriptive
 - Predictive

Classification

Introduction to Classification

- **Classification** uses information from multiple variables to allocate objects into two or more pre-specified classes.
- Unlike clustering, classification is a *supervised* learning method.
- Why do we classify objects?

Illustrative Example 1

- Suppose that you are working in marketing and would like to advertise new products.
- How could classification help you to perform this task?
 - *Assume that you have some client history.*

Linear Discriminant Analysis

Linear Discriminant Analysis

- **Linear Discriminant Analysis (LDA)** is a classification method based on eigenanalysis and is well suited for **multivariate normal data**.
- LDA identifies the combination of variables that best distinguishes sample units from different groups.
- LDA is a supervised learning approach and takes the group identity as the response variable.

LDA Procedure

- As combinations of variables can be more sensitive than individual variables, this approach combines the variables.
- Procedure:
 - 1 Compute the d -dimensional mean vectors for each of the k classes separately from the dataset.
 - 2 Compute the scatter matrices (between-class scatter matrix (SB) and within-class scatter matrix (SW)).
 - Max SB & Min SW .
 - 3 Calculate the eigenvectors and eigenvalues for the ratio of SW and SB .
 - 4 Sort the eigenvectors in descending order of their corresponding eigenvalues, and select the m eigenvectors with the largest eigenvalues to form a $d \times m$ -dimensional transformation matrix W .
 - 5 Use W to project the samples onto the new subspace.

LDA in R

- We can use the MASS package in R.
 - In Python.
- Usage:
 - `lda(grouping ~ X1 + X2 + ..., prior = proportions, CV = FALSE)`
 - If `CV = TRUE` returns results for leave-one-out cross-validation.
 - Can set `method = "t"` for more robust estimated based on a *t*-distribution (add `nu = df`).

Example 1

- Import the *Penguins2.csv* dataset and do the following:
 - 1 Use the code to scale the dataset.
 - 2 Perform some EDA on the dataset.
 - 3 Conduct LDA with the species as the response variable.
 - 4 Examine the results.
 - 5 Repeat the analysis but perform leave-one-out cross-validation.
- What do you notice?

Results information

- Prior probabilities of groups: defaults to the sample proportions found in the observed data.
- Group means: The multidimensional group means (*scaled values will be different than unscaled*).
- Coefficients of linear discriminants: Larger coefficients (either positive or negative) indicate variables that carry more weight with respect to that LD.
 - Note: There is one fewer LD than there were groups in the dataset (if you know that an observation doesn't belong to one of these groups then it must belong to the last group).
- Proportion of trace: The proportion of variation explained by each LD function (eigenvalue).
 - *Always sum to 1.*

LDA Plots

- We can visualize the results in a few different ways:
 - `plot(results)` gives a scatterplot of the linear discriminants and the class labels (may also use `pairs(results)`).
 - `plot(results, dimen = 1, type = "both")` gives histograms with densities of the LD1 for each group.
- You can see where the groups fall in the reduced dimensions.

Example 2

- Using your results from Example 1, do the following:
 - ① Generate a scatterplot of the linear discriminants and the class labels.
 - ② Generate a pairs plot of the linear discriminants and the class labels.
 - ③ Generate histograms with densities of the LD1 for each group.
- What do you notice?

LDA Predictions

- LDA may be used to predict the classes of unseen observations.
 - *How are the unseen values scaled?*
- We can use the `predict(model)` function to assess the training accuracy.
- We can use the `predict(model, new.data)` function to predict on unseen data.

Example 3

- Using the provided code, do the following:
 - ① Split your data into a training and testing set.
 - ② Run LDA on the training set.
 - ③ Evaluate the training accuracy of your model.
 - ④ Evaluate the accuracy of your model on the test set.
- What do you notice?

Comments on LDA

- LDA is an older method from the 1930's and may not be the best classifier available.
 - It should be useful for validating your clustering results.
- It is still commonly used in some domains and may serve as a useful baseline approach.
- You can change the prior probabilities in the modelling process depending on your objectives.
 - *What are the impacts of misclassifications?*

Next Lecture

- We will discuss more **Discriminant Analysis**.
- Quadratic discriminant analysis
- Distance-based discriminant analysis.

Exercise 1

- Import the wine dataset from the *HDclassif* package.
- Take some time to get to know the data (including visualizations).
- Create a 75/25 training and testing split.
 - Remember, any scaling needs to be done before this step.
- Perform LDA on your training set.
 - What is the training accuracy of your model?
- What is the testing accuracy of your model?
- How do these results compare to the clustering that you have done?

References & Resources

- 1 Bakker, J. D. (2026) Applied Multivariate Statistics in R
 - 2 Peng R. D. (2020) *Exploratory Data Analysis with R*
 - 3 Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
- LDA
 - `lda()`
 - `confusionMatrix()`